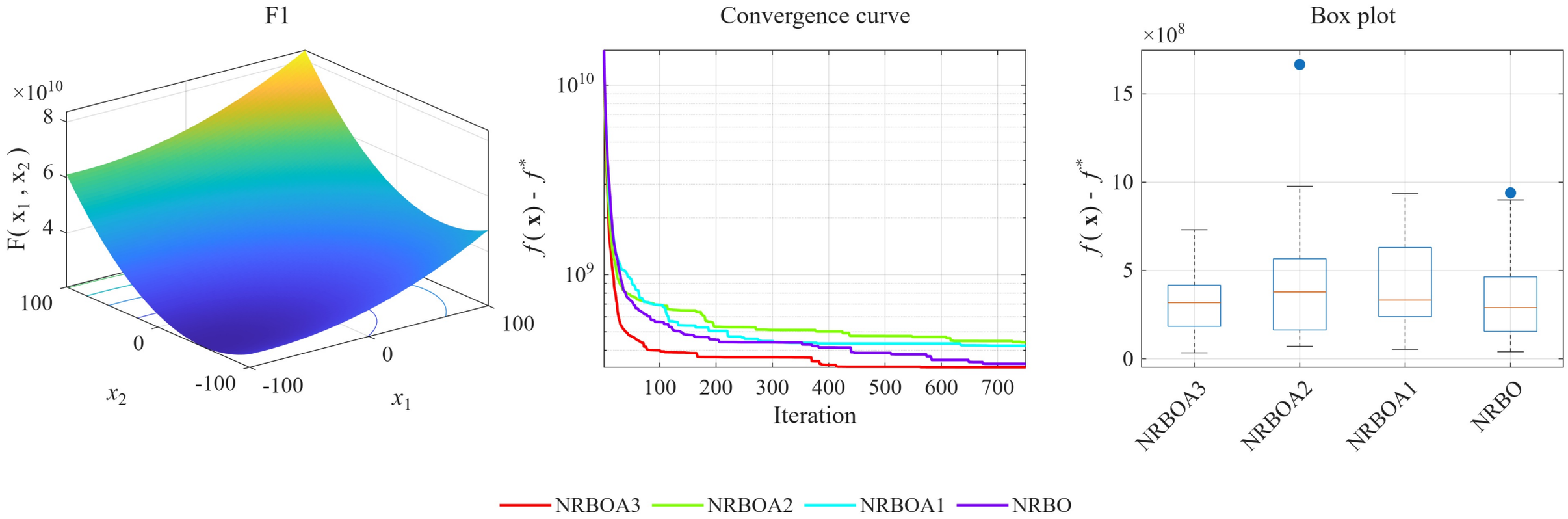

F1
×10^10
8
6
4
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^10
10^9
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^8
15
10
5
0
f(x) - f*
NRBOA3
NRBOA2
NRBOA1
NRBO
NRBOA3
NRBOA2
NRBOA1
NRBO